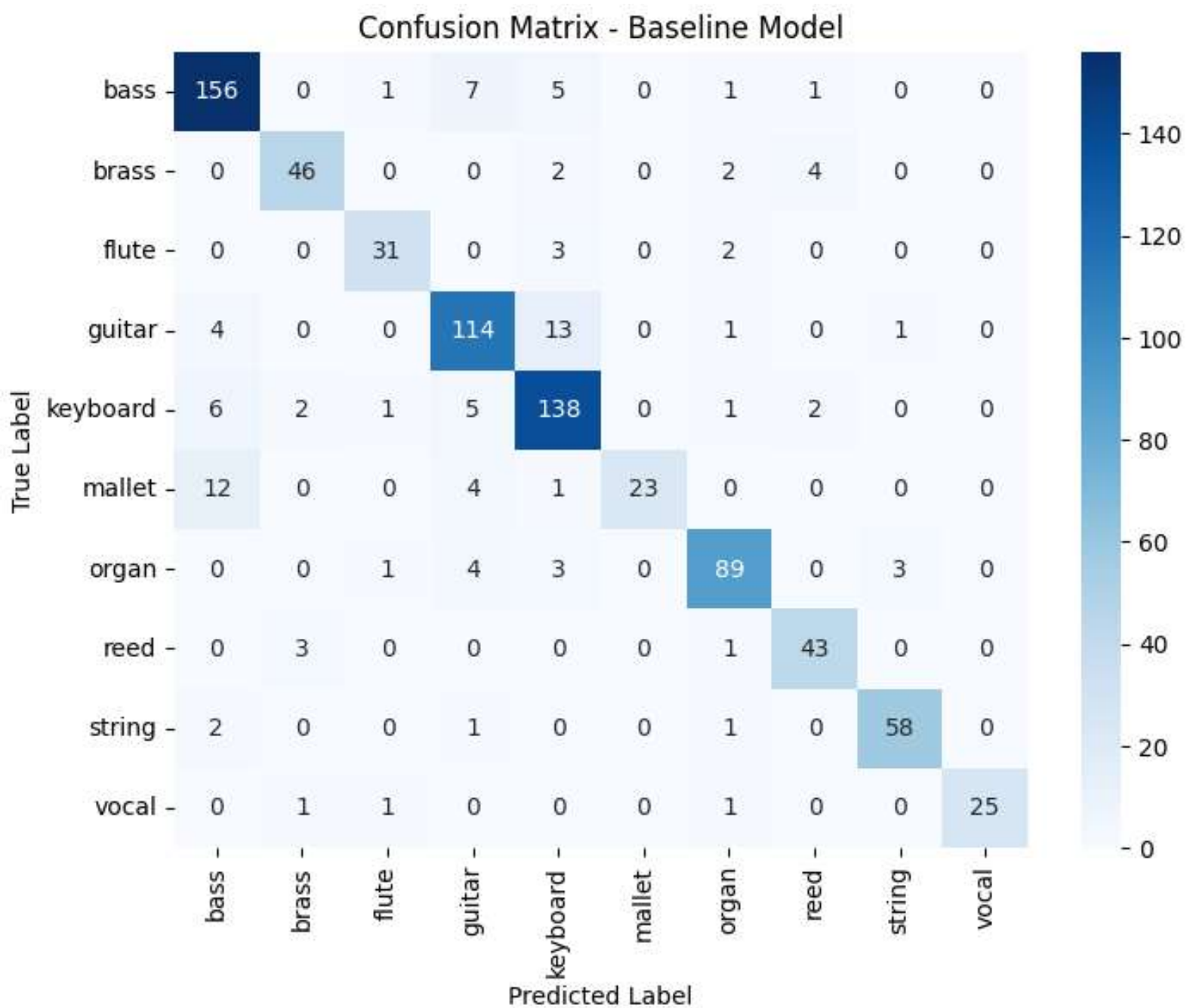


Classification Report (Baseline Model):

	precision	recall	f1-score	support
bass	0.87	0.91	0.89	171
brass	0.88	0.85	0.87	54
flute	0.89	0.86	0.87	36
guitar	0.84	0.86	0.85	133
keyboard	0.84	0.89	0.86	155
mallet	1.00	0.57	0.73	40
organ	0.90	0.89	0.89	100
reed	0.86	0.91	0.89	47
string	0.94	0.94	0.94	62
vocal	1.00	0.89	0.94	28
accuracy			0.88	826
macro avg	0.90	0.86	0.87	826
weighted avg	0.88	0.88	0.87	826



GridSearchCV



```
GridSearchCV(cv=3, estimator=RandomForestClassifier(random_state=42), n_jobs=-1,  
             param_grid={'max_depth': [None, 20, 40],  
                          'min_samples_split': [2, 5],  
                          'n_estimators': [100, 200]})
```

best_estimator_: RandomForestClassifier

```
RandomForestClassifier(max_depth=20, n_estimators=200, random_state=42)
```

RandomForestClassifier



```
RandomForestClassifier(max_depth=20, n_estimators=200, random_state=42)
```

Classification Report (Optimized Model):

	precision	recall	f1-score	support
bass	0.89	0.92	0.90	171
brass	0.90	0.83	0.87	54
flute	0.97	0.83	0.90	36
guitar	0.85	0.85	0.85	133
keyboard	0.82	0.90	0.86	155
mallet	0.96	0.62	0.76	40
organ	0.93	0.91	0.92	100
reed	0.83	0.91	0.87	47
string	0.94	0.95	0.94	62
vocal	1.00	0.89	0.94	28
accuracy			0.88	826
macro avg	0.91	0.86	0.88	826
weighted avg	0.89	0.88	0.88	826

